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Effects of hydrostatic compression and tension on silicon-vacancy centers in diamond

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ABSTRACT

Hydrostatic deformation is an effective approach for tuning the quantum properties of color centers in diamond, with significant implications for quantum sensing, computing, and communication. Compared to the widely studied nitrogen-vacancy (NV) centers, silicon-vacancy (SiV) centers exhibit more than a tenfold increase in coherent photon emission. In this work, we investigate the effects of hydrostatic pressure and tension on the SiV center in diamond using first-principles calculations with the r^{2}SCAN meta-GGA (Generalized Gradient Approximation) functional. We demonstrate that under hydrostatic tension corresponding to an isotropic expansion exceeding 4%, the SiV center undergoes spontaneous symmetry breaking from the inversion-symmetric D_{3d} structure to the asymmetric C_{3v} configuration, similar to that of the NV center. Within the hydrostatic compression and tension range corresponding to isotropic deformations of −8%–4%, the optical properties and hyperfine parameters of the SiV center change monotonically, indicating promising potential for pressure- or deformation-sensing applications. A microscopic explanation of these trends is provided from an electronic structure perspective. The r^{2}SCAN meta-GGA functional shows high accuracy in calculating hyperfine parameters, in agreement with experimental results. This study enhances our understanding of the optical properties and hyperfine interactions of SiV defects in diamond, laying the groundwork for their potential use in hydrostatic pressure or strain sensing applications.

The XV^y point defects in diamond, where X (e.g., N, Si, Ge, and Sn) substitutes carbon atoms near vacancies (V) and y denotes the charge state, are a key solid-state platform for quantum technology applications. The negatively charged nitrogen-vacancy (NV^-) center in diamond has found wide application in various quantum technologies, including quantum bits (qubits) for quantum computing, single-photon emitters (SPEs) for quantum communication, and nanoscale sensors in quantum metrology.[1–4] As shown in Fig. 1(a), the NV^- defect center is formed when a nitrogen atom substitutes for a carbon atom in the diamond lattice, adjacent to a carbon vacancy, resulting in a local C_{3v} symmetry.[5] Despite its many advantages, only a small fraction of the emitted photoluminescence resides at the zero-phonon line (ZPL), which is characterized by a low Debye-Waller factor of approximately 0.04.[6,7]

In contrast to the NV^- center, the silicon-vacancy (SiV^-) defect center exhibits significantly improved optical properties.[8,9] The SiV^- center concentrates nearly all of its photoluminescence into a narrow and intense ZPL, with a Debye-Waller factor of 0.8, substantially higher than that of the NV^- center.[10] The SiV^- defect center [Fig. 1(b)] forms when a larger silicon atom replaces carbon near a vacancy, shifting toward it and creating a split structure with D_{3d} symmetry.[11] The inversion symmetry of SiV^- eliminates a permanent

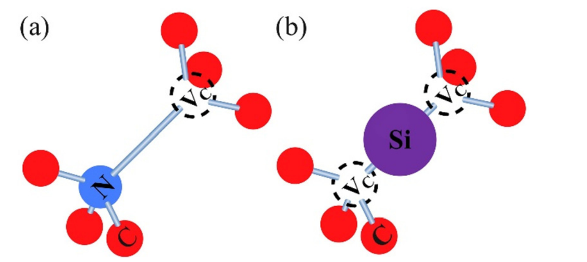

FIG. 1. Local atomic structures of (a) NV^- center and (b) SiV^- center in diamond.

dipole, ensuring spectral stability and reduced electric field sensitivity. Its response to lattice changes makes it ideal for high-precision temperature and pressure sensing.[12,13]

Hydrostatic pressure effectively induces isotropic lattice compression in the diamond lattice hosting SiV^- centers. In contrast, the influence of temperature on the ZPL is more intricate, arising from both lattice expansion/contraction and electron–phonon interactions. Recently, numerous studies have investigated pressure- and stress-mediated quantum sensing in SiV^- color center systems. For example, Vindolet *et al.* experimentally realized the modulation of the ZPL of SiV^- centers under hydrostatic pressure, reaching pressures as high as 180 GPa.[14,15] Theoretically, Mohseni *et al.* developed a theoretical framework for calculating hyperfine tensors in the presence of the Jahn-Teller effect, applying this approach to the SiV^- defect center. However, while the ZPL shift of SiV^- centers under hydrostatic pressure has been reported previously,[14,15] the behavior of the ZPL under hydrostatic tension has not yet been explicitly investigated. Of particular interest is the fact that experimental studies have demonstrated that nanodiamonds can endure tensile uniaxial strains of up to 9%.[16,17] This raises an important question regarding whether silicon-vacancy centers can remain stable under such significant tensile deformation.

One critical experimental technique for characterizing silicon-vacancy color centers in diamond is electron spin resonance (ESR). As early as 2002, Iakoubovskii *et al.* conducted studies on silicon-related defects in diamond.[18–21] Based on the experimental results reported by Iakoubovskii *et al.*, the KUL1 ($S = 1$, $g_1 = 2.0040$, $g_2 = 2.0035$, and $g_3 = 2.0035$) and KUL8 ($S = 1/2$, $g_1 = 2.00355$, $g_2 = 2.00338$, and $g_3 = 2.00338$) centers are defined as the neutral and negatively charged silicon-vacancy defects (SiV^0 and SiV^-), respectively. Edmonds *et al.* provided the hyperfine parameters for the KUL1 center, reporting values for one ^{29}Si atom ($A_\parallel = 66.2$ MHz, $A_\perp = 30.2$ MHz) and six equivalent ^{13}C atoms ($A_\parallel = 76.3$ MHz, $A_\perp = 78.9$ MHz).[22] In contrast, to date, there have been no reports on the hyperfine parameters of the KUL8 center.[18] In theoretical calculations, Gali *et al.* employed hybrid functionals to calculate the hyperfine parameters of SiV^0 centers.[23] The hyperfine parameters for carbon atoms match KUL1 center values closely, while those for the central silicon atom differ by ~23%, highlighting an incomplete understanding of SiV defect parameters. Stronger evidence is needed to assign silicon-vacancy defects to ESR-observed centers, and studying their evolution under external deformation or pressure is key for potential sensing applications.

In this study, we utilize the recently developed r^{2}SCAN meta-GGA functional to investigate the SiV center in diamond. Our primary focus is to explore how the geometry, optical properties, and hyperfine parameters of the SiV center evolve under hydrostatic compression and tension. Through this analysis, we aim to elucidate the microscopic origins of these changes, particularly the underlying modifications in the electronic structure.

We investigated the structural response of SiV defects in diamond under hydrostatic deformations ranging from −8% to +8%. Our results show that under hydrostatic compression, the SiV defect maintains its integrity, preserving the symmetry of the unstrained structure. However, when subjected to hydrostatic tension beyond 4%, significant structural changes occur. Figure 2 illustrates the spin density of both neutral and negatively charged SiV defects under 6% and 8% hydrostatic tension. At 6% hydrostatic tension, the SiV^0 defect no longer keeps the silicon atom centered between the two carbon vacancies. Instead, the silicon atom shifts toward one side, resulting in a three-coordinated silicon configuration. In this configuration, the Si−C bond length is 1.930 Å, and the distance between the silicon atom and the three remaining carbon atoms increases to 2.429 Å. At 8% hydrostatic tension, a similar deformation is observed, with the silicon atom still displaced toward the three adjacent carbon atoms. The Si−C bond length is 1.928 Å, and the distance to the remaining carbon atoms increases slightly to 2.588 Å. At 6% and 8% hydrostatic tension, the silicon atom adopts a three-coordinated configuration, but the displacement directions differ relative to the surrounding carbon framework (Fig. 2). The spin density is primarily localized on the more distant carbon atoms. This is confirmed by hyperfine parameter calculations, and the electron-nuclear spin coupling on these atoms is stronger than that on the closer carbon atoms (see the supplementary material).

This behavior arises from the larger atomic radius of silicon compared to carbon. Under hydrostatic compression, the contraction of the space around the carbon vacancy forces the silicon atom to remain between the two vacancies. As hydrostatic tension increases, the expanded local lattice accommodates the silicon atom, allowing the

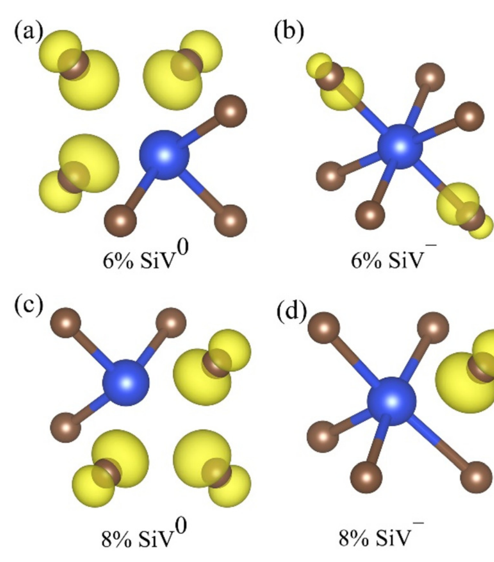

FIG. 2. Spin densities of SiV defects under hydrostatic tension: (a) neutral defect at 6%, (b) negatively charged defect at 6%, (c) neutral defect at 8%, and (d) negatively charged defect at 8%. Spin density is shown in yellow (0.02 e/Å^3 isosurface), with blue for silicon and brown for carbon atoms.

Si–C bond to relax. Given the Si–C bond length in SiC crystals is 1.89 Å,[24] the Si–C bond formed in the SiV defect gradually approaches this value under increasing hydrostatic tension, resembling the structure of the NV center, where nitrogen substitutes carbon and is adjacent to a vacancy.

At 6% hydrostatic tension, in the negatively charged SiV defect, two Si–C bond lengths are longer than the other four. Although the defect does not undergo significant reconstruction, small deformations persist, and the spin density is primarily located on the two more distant carbon atoms. At 8% hydrostatic tension, the negatively charged SiV defect undergoes substantial reconstruction, with the silicon atom bonding to five neighboring carbon atoms, forming a five-coordinated silicon configuration. The remaining carbon atoms are positioned farther away, with a distance of 2.519 Å. In this configuration, the spin density is mainly located on the isolated carbon atoms. At both 6% and 8% hydrostatic tension, the SiV defect no longer retains the symmetry of the undeformed state. Therefore, in contrast to the broader isotropic deformation range applied to defect-free diamond, we limit the hydrostatic deformation range in our study of SiV defects to −8% to +4%.

Figure 3(a) shows the variation of two charge transition levels (CTLs) associated with the SiV^- center, namely, (0/−) and (−/2−), as a function of deformation. When the Fermi level lies between these levels, the SiV^- center exists in a stable singly negatively charged state. Both CTLs change gradually across the range. The (0/−) level rises slightly with increasing tension, with a maximum variation under 0.1 eV, while the (−/2−) level decreases slightly, also within 0.1 eV. Based on these CTLs, we further analyze the ionization energy thresholds in Fig. 3(b). The ionization energy for the (−1 to −2) transition changes slowly with deformation, whereas the (−1 to 0) transition decreases significantly due to faster bandgap reduction under increasing tension (see the supplementary material). Throughout the range, the ZPL remains below both thresholds, confirming that SiV^- stays photostable under optical excitation. Although r^2SCAN underestimates the diamond bandgap, this does not significantly affect the computed CTLs, which rely on relative total energies rather than single-particle eigenvalues. Meta-GGA functionals have been shown to yield CTLs close to hybrid functional results, so the reported (0/−) and (−/2−) levels are reliable.[25–27]

At zero hydrostatic deformation, our calculated ZPL is 1.533 eV, which is in excellent agreement with the theoretical value reported by Maciaszek *et al.*[28] However, this calculated ZPL is slightly lower than the experimental value of 1.68 eV.[29] To reconcile this difference, we introduced a fixed offset, defined as the difference between the experimental and calculated values, and applied it to adjust the ZPL for all hydrostatic deformation conditions. The corrected ZPL values are presented in Fig. 3(b). Notably, for hydrostatic compression, the corrected ZPL aligns precisely with the experimental measurements reported by Vindolet *et al.*[14] Furthermore, we extended our calculations to include the monotonic variation of ZPL across a broader hydrostatic deformation range, from −8% to +4%. A linear fit of the corrected ZPL values reveals a slope of −1.265 eV/strain, with an adjusted R-squared value of 0.994, indicating a high degree of correlation and a robust fit to the data.

In an ideal isolated two-level system without electron–phonon coupling, the ZPL directly reflects the energy difference between the two states. To examine this relationship, we calculated the single-particle Kohn-Sham energy difference between the e_g and e_u levels and plotted this quantity as a function of hydrostatic deformation in Fig. 4(a). As expected, the energy gap decreases monotonically with increasing hydrostatic tension, which is consistent with the behavior of the ZPL. Notably, we observed that the e_u state undergoes a faster variation across the entire deformation range compared to the e_g state, indicating that the e_u state primarily governs the variation of the ZPL.

However, the SiV^- is not a simple isolated two-level system. It is embedded within the host diamond lattice, and as such, the electron excitation process involves complex electron–phonon coupling. To better describe the optical absorption and emission processes of point defects in solids, a more realistic approach is employed, using a one-dimensional configuration coordinate diagram as shown in Fig. 4(c), which takes static coupling effects into account.[30] Compared to the ZPL, E_{ab} is closer to the Kohn-Sham energy difference at all hydrostatic deformations, with the maximum difference between them being less than 0.05 eV at 4% hydrostatic tension. The difference between E_{ab} and ZPL represents the energy released due to lattice deformation induced by electron–phonon coupling during the excitation process. Furthermore, it is evident that this difference increases under hydrostatic tension and decreases under hydrostatic compression. The Q-value for the unstrained system is 0.14 $amu^{1/2}$ Å, while under −8% hydrostatic compression, the Q-value is 0.13 $amu^{1/2}$ Å, and under +4% hydrostatic tension, the Q-value increases to 0.18 $amu^{1/2}$ Å. Larger structural deformations correspond to greater energy differences between ZPL and the single-particle Kohn-Sham energy ($e_g - e_u$) in Fig. 4(a).

As shown in Fig. 4(b), both E_{ab} and E_{em} decrease monotonically with increasing hydrostatic deformation, following a trend consistent

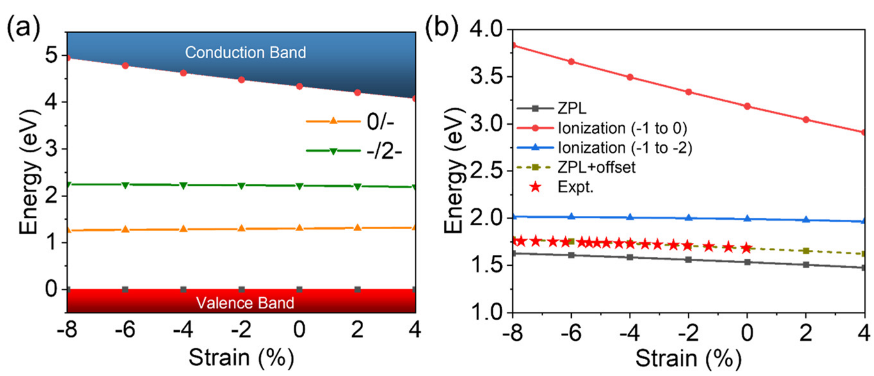

FIG. 3. (a) The (0/−) and (−/2−) CTLs of the SiV defect under deformation. (b) Ionization energies of SiV^-, including the (−1 to 0) and (−1 to −2) transitions, together with the ZPL energy (with and without a fixed offset). The offset ZPL closely matches experimental values measured under pressure.[14]

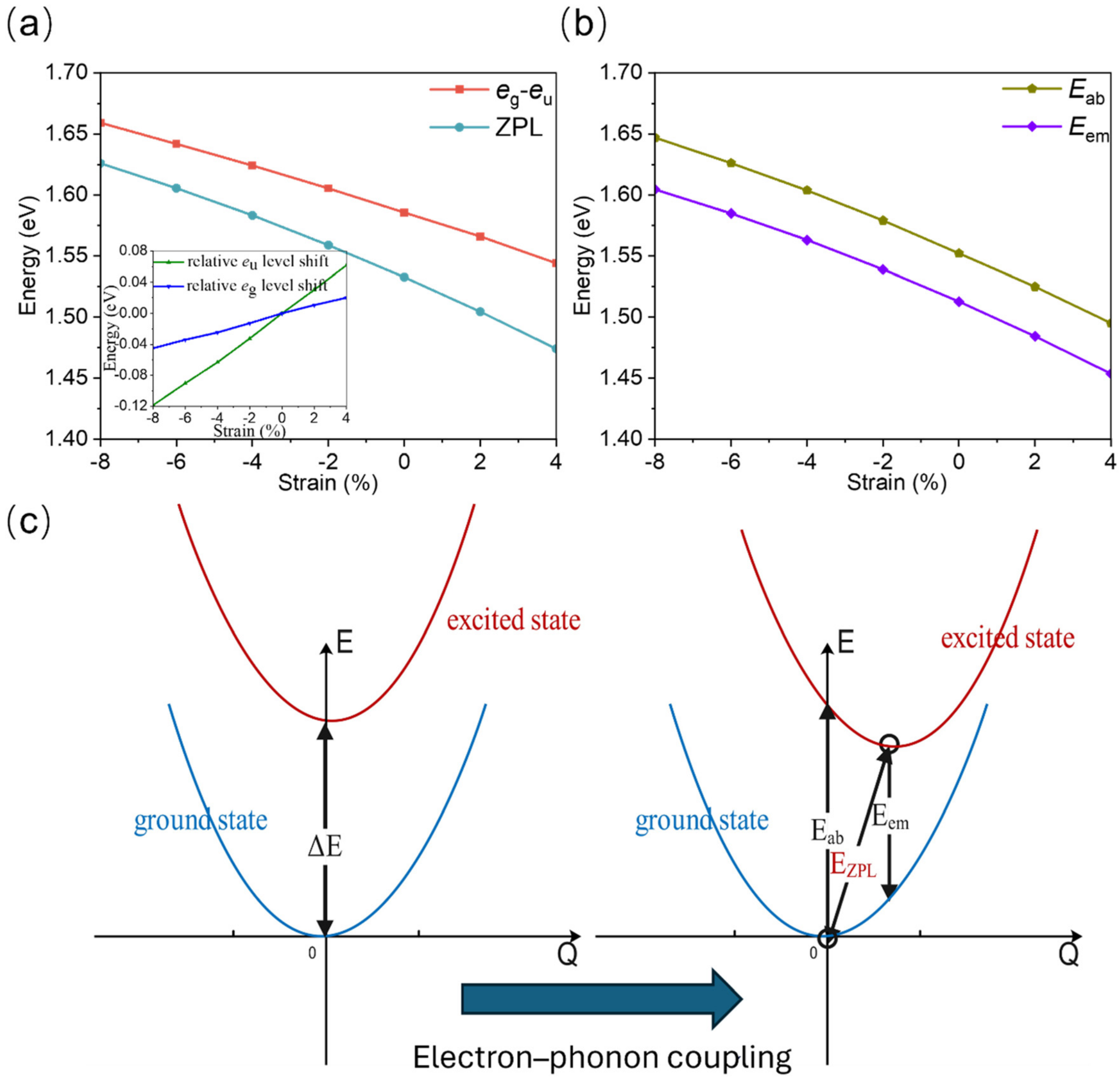

FIG. 4. (a) Variation of the energy difference between the single-particle Kohn-Sham levels (e_g and e_u) with deformation, showing a trend similar to the ZPL. The inset shows the single-particle Kohn-Sham levels (e_g and e_u) relative to the valence band maximum, with energies aligned to zero in the deformation-free case. (b) Absorption and emission energies of SiV^- under different deformations. (c) Configuration coordinate diagram illustrating the evolution from the unperturbed defect system to the system including electron–phonon coupling.

with the ZPL variation. This indicates that, in addition to the ZPL, both E_{ab} and E_{em} are also sensitive to hydrostatic deformation and can serve as effective monitoring indicators. Due to relaxations in both the excited and ground states, the emission energy is lower than the absorption energy. The Stokes shift, defined as the energy difference between absorption and emission,[31] was calculated and found to be almost independent of hydrostatic deformation, with an average value of 0.041 eV and a standard deviation of 0.001 eV. Consequently, the absorption-emission energy difference remains nearly constant over the entire hydrostatic deformation range.

In the unstrained case, the computed radiative lifetime was 13.67 ns, which closely aligns with the result (12.13 ns) reported by Thiering *et al.* using the HSE06 method.[32] Furthermore, our calculations reveal that the radiative lifetime decreases with increasing hydrostatic deformation (see the supplementary material). Compared to the ZPL energy, the change in the transition dipole moment has a more pronounced impact on the lifetime. With increasing hydrostatic deformation, the square of the transition dipole moment increases, leading to a reduction in the radiative lifetime.

The hyperfine parameters of the SiV center computed with the r^2SCAN functional show excellent agreement with experimental measurements, demonstrating markedly higher accuracy compared to HSE06 (see the supplementary material). We also calculated the hyperfine parameters, including $A_\parallel$, $A_\perp$, f, and d, for both SiV^0 and SiV^- under various hydrostatic deformation conditions. These parameters were calculated for both the central silicon atom and the nearest-neighbor carbon atoms. It is important to note that our calculations explicitly account for core contributions. Previous studies have demonstrated that incorporating core contributions leads to results that align more closely with experimental measurements.[23,33] However, we observed an anomalous behavior in the hyperfine parameter of the central silicon atom at −4% hydrostatic compression, where the value deviates significantly from those at other deformation levels. Upon further investigation, we determined that the source of this discrepancy was an unexpected enhancement of the core contributions of the silicon atom at this hydrostatic compression level. Consequently, in the main text, we focus only on reporting the variations in the hyperfine parameters for the carbon atoms under hydrostatic deformation.

For both charge states, all four parameters exhibit a monotonic increase with increasing hydrostatic deformation, as shown in Fig. 5. Among these, $A_{\parallel}$ shows the most rapid increase, while d increases at the slowest rate. The mathematical expressions for the isotropic part (f) and the anisotropic part (d) are as follows:[34–36]

$$f = \frac{8\pi}{3}\frac{\mu_0}{4\pi} g_e \mu_B g_n \mu_n |c_s|^2 \eta |\phi_s(0)|^2 = 3777 \times (1 - |c_p|^2)\eta \,\mathrm{MHz}, \quad (1)$$

$$d = \frac{2}{5}\frac{\mu_0}{4\pi} g_e \mu_B g_n \mu_n |c_p|^2 \eta \left\langle \phi_p \left| \frac{1}{r^3} \right| \phi_p \right\rangle = 107.4 \times |c_p|^2 \eta \,\mathrm{MHz}. \quad (2)$$

Here, $|\phi_s(0)|^2$ represents the probability density of the s orbital at the position of the ^{13}C nucleus, and $\langle \phi_p | \frac{1}{r^3} | \phi_p \rangle$ is the average $\frac{1}{r^3}$ value of the ϕ_p, which are determined numerically and available in published studies.[37] The symbol η denotes the electron spin density at the nucleus, g_n is the nuclear g factor of ^{13}C, and μ_N is the nuclear magneton. These relationships enable the determination of hydrostatic deformation-dependent changes in the orbital hybridization ratio $\chi = |c_s|^2/|c_p|^2$ and the spin density η. As illustrated in Fig. 5, both η and χ exhibit an increase with increasing deformation. The concurrent rise of η and χ facilitates the enhancement of f, leading to a more rapid increase in f under deformation. By contrast, the increase in χ is associated with a reduction in $|c_p|^2$. While η continues to grow with deformation, the decrease in $|c_p|^2$ counteracts the growth of d, resulting in a slower increase in d.

For the unstrained case, our calculated A_{PLE} is −36.95 MHz, which is in close agreement in magnitude with the experimental value of 35 MHz reported in the literature,[38] indicating good consistency. Additionally, we observe that the hyperfine interaction in the excited state is smaller than that in the ground state. This can be explained by symmetry considerations: in the ground state with $e_u^4 e_g^3$ configuration, the unpaired electron occupies the e_g orbital, which has inversion symmetry, whereas in the excited state with $e_u^3 e_g^4$ configuration, the unpaired electron occupies the e_u orbital, which has inversion antisymmetry. The node in the excited state orbital results in reduced spin density at the nuclear spin location, leading to a smaller hyperfine interaction in the excited state.

The hyperfine parameters of the ground state are consistently larger than those of the excited state at the same hydrostatic deformation. As the deformation increases, the ground state hyperfine parameters rise rapidly, while those of the excited state decrease gradually. As a result, the hydrostatic-deformation dependence of A_{PLE} is dominated by the ground-state contribution. Consequently, A_{PLE} decreases monotonically with increasing deformation. A linear fit yields a high adjusted R-squared value of 0.995, with a slope of −233.57 MHz/strain, indicating that A_{PLE} can serve as an effective probe of hydrostatic deformation.

In this study, we utilized first-principles calculations to explore the response of SiV centers to hydrostatic deformation. Our results

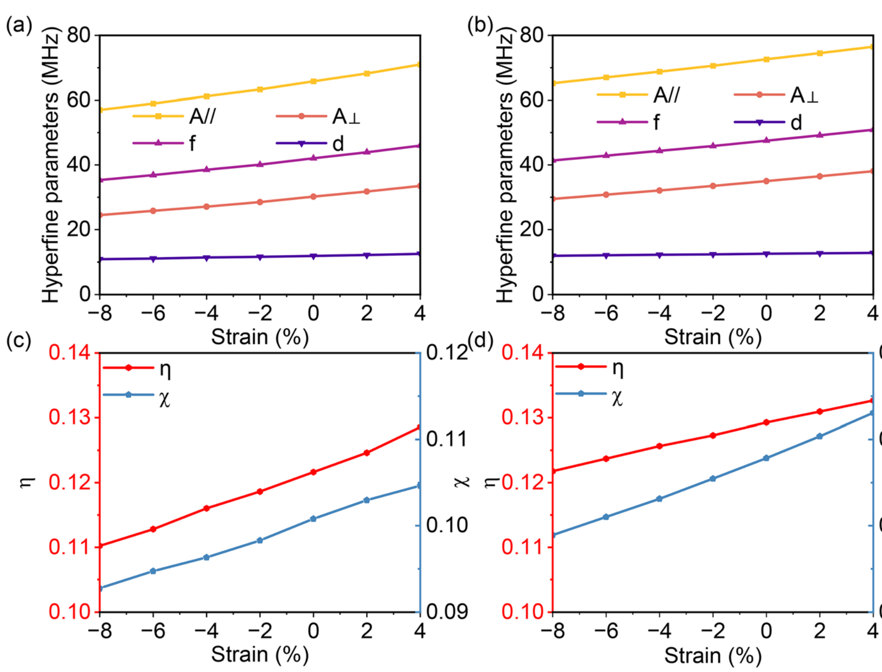

FIG. 5. Variation of the hyperfine parameters and electron spin density (η and χ) of SiV^0 and SiV^- under hydrostatic deformation: (a) SiV^0 hyperfine parameters, (b) SiV^- hyperfine parameters, (c) SiV^0 η and χ, and (d) SiV^- η and χ.

indicate that while pristine diamond can sustain isotropic deformations from −8% to 8%, the SiV center remains stable only within the range of −8%–4%; beyond this range, the center undergoes pronounced structural distortion. Regarding optical properties, the ZPL, E_{ab}, and E_{em} decrease monotonically with increasing deformation, suggesting their potential as reliable sensing indicators. Moreover, the radiative lifetime is also found to decrease as deformation increases. For hyperfine parameters, $A_{\parallel}$, $A_{\perp}$, and f show a significant increase, while d varies only slightly. Meanwhile, A_{PLE} decreases monotonically and can be accessed optically. These trends were analyzed from an electronic structure perspective, providing a microscopic understanding of the observed behavior.

See the supplementary material for computational details and additional results on the electronic, optical, and hyperfine properties of SiV centers under hydrostatic deformation.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yunliang Yue: Conceptualization (equal); Data curation (lead); Formal analysis (equal); Investigation (equal); Software (lead); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Min Wang:** Investigation (supporting). **Yaxuan Liu:** Investigation (supporting). **Runxi Guo:** Investigation (supporting). **Han Zhang:** Investigation (supporting). **Huamu Xie:** Conceptualization (equal). **Yee Sin Ang:** Formal analysis (equal); Funding acquisition (equal); Supervision (equal). **Shibo Fang:** Conceptualization (equal); Visualization (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its supplementary material.

REFERENCES

[1] R. Schirhagl, K. Chang, M. Loretz *et al.*, Annu. Rev. Phys. Chem. **65**, 83–105 (2014).
[2] Á. Gali, Nanophotonics **8**, 1907–1943 (2019).
[3] S. Pezzagna and J. Meijer, Appl. Phys. Rev. **8**, 011308 (2021).
[4] K. O. Ho, K. C. Wong, M. Y. Leung *et al.*, J. Appl. Phys. **129**, 241101 (2021).
[5] L. Razinkovas, M. Maciaszek, F. Reinhard *et al.*, Phys. Rev. B **104**, 235301 (2021).
[6] G. Zhang, Y. Cheng, J.-P. Chou *et al.*, Appl. Phys. Rev. **7**, 031308 (2020).
[7] I. Aharonovich, S. Castelletto, D. A. Simpson *et al.*, Rep. Prog. Phys. **74**, 076501 (2011).
[8] U. F. D'Haenens-Johansson, A. Edmonds, B. Green *et al.*, Phys. Rev. B **84**, 245208 (2011).
[9] A. Petrone, R. A. Beck, J. M. Kasper *et al.*, Phys. Rev. B **98**, 205405 (2018).
[10] E. Neu, D. Steinmetz, J. Riedrich-Möller *et al.*, New J. Phys. **13**, 025012 (2011).
[11] L. J. Rogers, K. D. Jahnke, M. W. Doherty *et al.*, Phys. Rev. B **89**, 235101 (2014).
[12] A. A. Razgulov, S. G. Lyapin, A. P. Novikov *et al.*, Phys. Rev. B **111**, 024114 (2025).
[13] S. Revesz, A. Misiara, J. B. S. Abraham *et al.*, ACS Photonics **11**, 2221–2227 (2024).
[14] B. Vindolet, M.-P. Adam, L. Toraille *et al.*, Phys. Rev. B **106**, 214109 (2022).
[15] M. Mohseni, L. Razinkovas, V. Žalandauskas *et al.*, Phys. Rev. B **112**, 155201 (2025).
[16] A. Banerjee, D. Bernoulli, H. Zhang *et al.*, Science **360**, 300–302 (2018).
[17] C. Dang, J.-P. Chou, B. Dai *et al.*, Science **371**, 76–78 (2021).
[18] K. Iakoubovskii and A. Stesmans, Phys. Rev. B **66**, 195207 (2002).
[19] K. Iakoubovskii and A. Stesmans, J. Phys.: Condens. Matter **14**, R467 (2002).
[20] K. Iakoubovskii, A. Stesmans, M. Nesladek *et al.*, phys. status solidi **193**, 448–456 (2002).
[21] K. Iakoubovskii, A. Stesmans, K. Suzuki *et al.*, Diamond Relat. Mater. **12**, 511–515 (2003).
[22] A. Edmonds, M. E. Newton, P. Martineau *et al.*, Phys. Rev. B **77**, 245205 (2008).
[23] A. Gali and J. R. Maze, Phys. Rev. B **88**, 235205 (2013).
[24] J. P. Rino, I. Ebbsjö, P. S. Branicio *et al.*, Phys. Rev. B **70**, 045207 (2004).
[25] D. Wickramaratne and J. L. Lyons, Phys. Rev. B **109**, 245201 (2024).
[26] T. Rauch, F. Munoz, M. A. Marques *et al.*, Phys. Rev. B **104**, 064105 (2021).
[27] G. Abbas, O. Bulancea-Lindvall, J. Davidsson *et al.*, Appl. Phys. Lett. **126**, 154001 (2025).
[28] M. Maciaszek, V. Žalandauskas, R. Silkinis *et al.*, J. Chem. Phys. **159**, 084708 (2023).
[29] C. D. Clark, H. Kanda, I. Kiflawi *et al.*, Phys. Rev. B **51**, 16681–16688 (1995).
[30] A. Alkauskas, Q. Yan, and C. G. Van de Walle, Phys. Rev. B **90**, 075202 (2014).
[31] A. Alkauskas, M. D. McCluskey, and C. G. Van de Walle, J. Appl. Phys. **119**, 181101 (2016).
[32] G. Thiering and A. Gali, Phys. Rev. X **8**, 021063 (2018).
[33] R. Kuate Defo, E. Kaxiras, and S. L. Richardson, Phys. Rev. B **104**, 075158 (2021).
[34] J. Loubser and J. van Wyk, Rep. Prog. Phys. **41**, 1201 (1978).
[35] X.-F. He, N. B. Manson, and P. T. Fisk, Phys. Rev. B **47**, 8816 (1993).
[36] M. Barson, P. Reddy, S. Yang *et al.*, Phys. Rev. B **99**, 094101 (2019).
[37] J. Morton and K. Preston, J. Magn. Reson. **30**, 577–582 (1978).
[38] L. J. Rogers, K. D. Jahnke, M. H. Metsch *et al.*, Phys. Rev. Lett. **113**, 263602 (2014).